**Collections Assignment**

**Stack Program**

Card.java:

```
public class Card {
  String name;
  String suit;

  public Card (String name, String suit) {
    this.name = name;
    this.suit = suit;
  }

  public String toString() {
    return name + " of " + suit;
  }
}
```

mainProgram.java:

```
import java.util.Random;
import java.util.Stack;

public class mainProgram {

  public static void main(String[] args) {

    // Deck of 14 cards
    Card[] deck = new Card[14];

    String[] names = {
        "One", "Two", "Three", "Four", "Five",
        "Six", "Seven", "Eight", "Nine", "Ten",
        "Jack", "Queen", "King", "Ace"
    };

    String[] suits = { "Hearts", "Spades", "Diamonds", "Clubs" };

    for (int i = 0; i < deck.length; i++) {
```

```
        deck[i] = new Card(names[i], suits[i % 4]);
    }

    // Create stack
    Stack<Card> stack = new Stack<>();
    Random random = new Random();

    // Push 5 random cards
    System.out.println("Pushing cards:\n");
    for (int i = 0; i < 5; i++) {
        Card randomCard = deck[random.nextInt(deck.length)];
        stack.push(randomCard);
        System.out.println(randomCard);
    }

    // Pop cards from stack
    System.out.println("\nPopping cards:\n");
    while (!stack.isEmpty()) {
        System.out.println(stack.pop());
    }
  }
}
```

Output:

```
Pushing cards:

Four of Clubs
Nine of Hearts
Seven of Diamonds
Nine of Hearts
Eight of Clubs

Popping cards:

Eight of Clubs
Nine of Hearts
Seven of Diamonds
Nine of Hearts
Four of Clubs

Process finished with exit code 0
```

**Queue and ArrayList Program**

Person.java:

```
public class Person {
  String name;
  int ticketNumber;

  public Person (String name) {
    this.name = name;
  }
}
```

mainProgram.java:

```
import java.util.*;

public class mainProgram {
  public static void main(String[] args) {

    Queue<Person> line = new LinkedList<>();
    ArrayList<Integer> tickets = new ArrayList<>();
    Random random = new Random();

    ArrayList<String> names = new ArrayList<>();
    names.add("Bruna");
    names.add("Isabela");
    names.add("Viviane");
    names.add("Rachel");
    names.add("Sergio");
    names.add("Rodrigo");
    names.add("Romulo");

    Collections.shuffle(names); // random order

    for (String n : names) {
      line.add(new Person(n));
    }

    for (int i = 0; i < 7; i++) {
```

```
        tickets.add(random.nextInt(1000));
      }

      int i = 0;
      while (!line.isEmpty()) {
        Person p = line.remove();
        p.ticketNumber = tickets.get(i);
        System.out.println(p.name + " got ticket " + p.ticketNumber);
        i++;
      }
    }
}
```

Output:

```
Romulo got ticket 626
Bruna got ticket 519
Rachel got ticket 35
Sergio got ticket 512
Isabela got ticket 935
Viviane got ticket 153
Rodrigo got ticket 634

Process finished with exit code 0
```

**Map Program**

Adress.java:

```
public class Address {
  private String firstName;
  private String lastName;
  private String street;

  public Address(String firstName, String lastName, String street) {
    this.firstName = firstName;
    this.lastName = lastName;
    this.street = street;
```

```
  }

  public String getLastName() {
    return lastName;
  }

  @Override
  public String toString() {
    return firstName + " " + lastName + " - " + street;
  }
}
```

AddressList.java:

```
import java.util.ArrayList;
import java.util.Collection;
import java.util.HashMap;
import java.util.Map;

public class AddressBook {

  private Map<String, ArrayList<Address>> book = new HashMap<>();

  public void add(Address a) {
    book.putIfAbsent(a.getLastName(), new ArrayList<>());
    book.get(a.getLastName()).add(a);
  }

  public boolean remove(String lastName) {
    return book.remove(lastName) != null;
  }

  public Collection<Address> find(String lastName) {
    return book.get(lastName);
  }

  public int count() {
    int total = 0;
    for (ArrayList<Address> list : book.values()) {
```

```
        total += list.size();
      }
      return total;
    }

    public Collection<Address> getAll() {
      ArrayList<Address> all = new ArrayList<>();
      for (ArrayList<Address> list : book.values()) {
        all.addAll(list);
      }
      return all;
    }
}
```

ConsoleIO.java:

```
import java.util.Scanner;

public class ConsoleIO {
  private Scanner sc = new Scanner(System.in);

  public void print(String message) {
    System.out.println(message);
  }

  public String readString(String message) {
    System.out.print(message);
    return sc.nextLine();
  }
}
```

AddressBookController.java:

```
import java.util.Collection;

public class AddressBookController {

  public static void main(String[] args) {

    ConsoleIO io = new ConsoleIO();
```

```
AddressBook book = new AddressBook();

boolean running = true;

while (running) {
   io.print("\nPlease select the operation you wish to perform:\n");
   io.print("1) Add Address");
   io.print("2) Delete Address");
   io.print("3) Find Address");
   io.print("4) List Address Count");
   io.print("5) List All Addresses");
   io.print("6) Exit");

   String choice = io.readString("\nEnter choice: ");

   if (choice.equals("1")) {
      io.print("\nAdd Address Menu:\n");

      String first = io.readString("Please enter first name: ");
      String last = io.readString("Please enter last name: ");
      String street = io.readString("Please enter street address: ");

      book.add(new Address(first, last, street));
      io.print("\nAddress Added. Press Enter to go to Main Menu.");
      io.readString("");

   } else if (choice.equals("2")) {
      String last = io.readString("Enter last name to delete: ");
      boolean removed = book.remove(last);

      if (removed) io.print("Addresses deleted.");
      else io.print("Address not found.");

   } else if (choice.equals("3")) {
      String last = io.readString("Enter last name to find: ");
      Collection<Address> found = book.find(last);

      if (found != null) {
```

```
            for (Address a : found) {
                io.print(a.toString());
            }
        } else {
            io.print("Address not found.");
        }

    } else if (choice.equals("4")) {
        io.print("Address Count: " + book.count());

    } else if (choice.equals("5")) {
        io.print("\nAll Addresses:");
        for (Address a : book.getAll()) {
            io.print(a.toString());
        }

    } else if (choice.equals("6")) {
        running = false;
        io.print("Goodbye!");
    }
  }
 }
}
```

Output:

```
Please select the operation you wish to perform:

1) Add Address
2) Delete Address
3) Find Address
4) List Address Count
5) List All Addresses
6) Exit

Enter choice: 1

Add Address Menu:

Please enter first name: Valencia
Please enter last name: College
Please enter street address: 1800 S. Kirkman Rd., Orlando, FL 32811

Address Added. Press Enter to go to Main Menu.
```

```
Please select the operation you wish to perform:

1) Add Address
2) Delete Address
3) Find Address
4) List Address Count
5) List All Addresses
6) Exit

Enter choice: 3
Enter last name to find: College
Valencia College - 1800 S. Kirkman Rd., Orlando, FL 32811

Please select the operation you wish to perform:

1) Add Address
2) Delete Address
3) Find Address
4) List Address Count
5) List All Addresses
6) Exit

Enter choice: 5

All Addresses:
Valencia College - 1800 S. Kirkman Rd., Orlando, FL 32811
```